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Degree/Course: BTECH COMPUTER SCIENCE AND ENGINEERING

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Title of the Work: Sentimental Analysis Using Social Media Comments

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ACKNOWLEDGEMENT

We express our humble gratitude to **Dr. C. Muthamizhchelvan**, Vice-Chancellor, SRM Institute of Science and Technology, for the facilities extended for the project work and his continued support.

We extend our sincere thanks to **Dr. Leenus Jesu Martin M**, Dean - CET, SRM Institute of Science and Technology, for his invaluable support.

We wish to thank **Dr. Revathi Venkataraman**, Professor and Chairperson, School of Computing, SRM Institute of Science and Technology, for her support throughout the project work.

We encompass our sincere thanks to, **Dr. M. Pushpalatha**, Professor and Associate Chairperson - CS, School of Computing and **Dr. Lakshmi**, Professor and Associate Chairperson - AI, School of Computing, SRM Institute of Science and Technology, for their invaluable support.

We are incredibly grateful to our Head of the Department, **Dr. G Niranjana**, Professor & Department of Computing Technologies, SRM Institute of Science and Technology, for her suggestions and encouragement at all the stages of the project work.

We want to convey our thanks to our Project Coordinators, Panel Head, and Panel Members Department of Computing Technologies, SRM Institute of Science and Technology, for their inputs during the project reviews and support.

We register our immeasurable thanks to our Faculty Advisor, **Dr. S Ramesh**, Department of Computing Technologies, SRM Institute of Science and Technology, for leading and helping us to complete our course.

Our inexpressible respect and thanks to our guide, **Dr. M. Murali**, Department of Computing Technologies, SRM Institute of Science and Technology, for providing us with an opportunity to pursue our project under her mentorship. She provided us with the freedom and support to explore the research topics of our interest. Her passion for solving problems and making a difference in the world has always been inspiring.

We sincerely thank all the staff members of Department of Computing Technologies, School of Computing, S.R.M Institute of Science and Technology, for their help during our project. Finally, we would like to thank our parents, family members, and friends for their unconditional love, constant support and encouragement.



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BONAFIDE CERTIFICATE

Certified that 21CSP302L – Project report titled **“SENTIMENTAL ANALYSIS USING SOCIAL MEDIA COMMENTS”** is the bonafide work of **“AMRITANSHU SHIWANSHI [RA2211003010756] and N SATYA DINESH [RA2211003010795]”** who carried out the project work under our supervision. Certified further, that to the best of my knowledge the work reported herein does not form any other project report or dissertation based on which degree or award was conferred on an earlier occasion on this or any other candidate.

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Examiner I

Examiner II

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ABSTRACT

The research investigates the use of machine learning algorithms for the automation of sentiment classification from social media comments, which will help detect and analyze public opinion early enough to assist decision-making processes across different domains. The research focuses on an exhaustive comparative evaluation of three widely used machine learning architectures, namely Logistic Regression, Linear Support Vector Classifier (LinearSVC), and Random Forest, for the classification of three major sentiment classes: positive, negative, and neutral. The dataset applied for this work was obtained from Reddit, featuring thousands of user comments, and it had been organized into a clean CSV format with proper labeling, thereby aiding reproducibility and minimizing annotation errors. In addition, the dataset was divided into training, validation, and testing subsets in such a way that a stratified approach maintains class balance during evaluation. Each model was implemented once with default parameters and once with hyperparameter tuning, thus enabling the impact of algorithmic choices and training strategies to be rigorously evaluated. Preprocessing techniques, including text normalization, stopwords removal, and TF-IDF vectorization, were performed to improve generalizability and robustness. All models were then assessed by means of accuracy, precision, recall, and F1-score, and confusion matrices were obtained for detailed error analysis. The results indicated that default LinearSVC produced the overall best accuracy with the highest value of 94.12%, giving the lowest class-wise imbalance performance, whereas tuned versions, especially tuned Random Forest, failed to improve and sometimes resulted in overfitting due to unsuitable hyperparameter tuning schemes. Of note, distinguishing neutral comments constituted the most challenging task across models, requiring specific class balancing techniques or improved text feature engineering methods. The project also emphasizes that standardized CSV organization with clean labeling gives better performance and fewer errors than any loosely structured datasets. The deductions made herein elaborate the way to determine successful automated sentiment analysis systems being based on scrutiny of model selection, hyperparameter control, and standardized dataset handling. The way is now clear for more studies on robust language models, imbalance-aware training, and real-world deployment across domains like marketing, mental health, and policy-making, thus adding further to the strides already being made in sentiment analysis from social media data.

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ABBREVIATIONS

AI – Artificial Intelligence

NLP – Natural Language Processing

BERT – Bidirectional Encoder Representations from Transformers

RoBERTa – Robustly Optimized BERT Approach

TF-IDF – Term Frequency-Inverse Document Frequency

EDA – Exploratory Data Analysis

SVR – Support Vector Regression

SDG – Sustainable Development Goals

WHO – World Health Organization

SMOTE – Synthetic Minority Over-sampling Technique

NLTK – Natural Language Toolkit

ROC – Receiver Operating Characteristic

CPU – Central Processing Unit

GPU – Graphics Processing Unit

Adam – Adaptive Moment Estimation (Optimizer)